



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Adaptive Software Engineering – (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	MedFlow AI – An Agentic Cloud-Native Healthcare Intelligence Platform using Microservices, DevOps and DevSecOps	
S. No.	University ID	Name
1	2420030586	JITHYASRI SAMMETA
2	2420030474	DOMA AKSHAYA
3	2420030267	NALLAPU SREEJA
4		
Name of the Guide		DR.ARCHANA KALIDINDI

Abstract

Healthcare organizations increasingly rely on digital systems to manage patients, clinicians, appointments, medical records, laboratory services, pharmacy operations, billing, and emergency care. Yet many conventional healthcare management applications remain fragmented and process-centric, demanding heavy manual coordination across departments while offering limited intelligent support. The result is duplicated information, delayed communication, inefficient appointment and queue management, difficulty interpreting medical documents, fragmented patient records, and slow response to operational and emergency needs — underscoring the need for a scalable platform that integrates healthcare services with secure, context-aware, intelligent assistance.

MedFlow AI addresses this gap through a service-oriented microservices architecture combined with artificial intelligence and modern software engineering practice. The platform delivers role-based workflows for patients, doctors, nurses, receptionists, laboratory staff, pharmacists, billing personnel, emergency staff, and administrators, covering authentication, patient profiles and histories, appointments, queue handling, medical records, lab reports, prescriptions, pharmacy operations, billing, notifications, and emergency support. An AI service built with Spring AI and the Google Gemini API coordinates six specialized agents — Patient, Doctor, Emergency, Pharmacy, Billing, and Hospital Operations Assistants — each capable of multi-step reasoning and tool use within its domain rather than single-shot Q&A: summarizing medical reports, organizing clinical information, guiding appointments and queues, surfacing medication and inventory insights, explaining bills, and recommending operational actions. All agents function strictly as decision-support tools operating under human oversight, not as autonomous diagnostic or treatment systems.

The system is implemented using Java 21, Spring Boot/Spring Cloud, React.js, and PostgreSQL with a database-per-service architecture. Security and communication are handled through Spring Security/JWT, API Gateway, Eureka, and OpenFeign. Docker, Kubernetes, and AWS support cloud-native deployment, while Jira, GitHub Actions, SonarQube, Trivy, OWASP ZAP, Prometheus, and Grafana enable Agile development, CI/CD, DevSecOps, and monitoring. Together, these technologies provide a scalable, secure, and reliable platform that integrates agentic AI with modern cloud-native software engineering.

Signature of the Guide

HOD